BCA701 – DEEP LEARNING
MODULE 3 – DETAILED STUDY MATERIAL

VTU 2022 Scheme | Regularization for Deep Learning and Optimization for Training Deep Models

Coverage: Overfitting, regularization, L1 and L2 parameter norm penalties, role of λ, weight decay, data augmentation, semi-supervised learning, gradient descent, ill-conditioning, local minima, plateaus, saddle points, flat regions, solved numericals, formula sheet, and exam preparation.

1. Module Overview

Module 3 has two connected parts: **Regularization for Deep Learning** and **Optimization for Training Deep Models**. Regularization improves generalization by preventing a model from fitting noise or irrelevant patterns. Optimization concerns the process of finding weights that minimize the loss function during training.

The key prescribed optimization challenges are: **ill-conditioning, local minima, plateaus, saddle points, and other flat regions**.

2. Overfitting

2.1 Definition

Overfitting occurs when a neural network learns the training data too well, including noise, accidental correlations, and unnecessary details, but performs poorly on new or unseen examples.

2.2 Typical sign

Training result	Validation/test result	Interpretation
High accuracy	High accuracy	Good generalization
High accuracy	Much lower accuracy	Overfitting
Low accuracy	Low accuracy	Underfitting

2.3 Example and analogy

A student who memorizes the answers to old question papers but cannot solve a newly framed question is analogous to an overfitted model. Similarly, an image classifier may memorize background details in training images instead of learning useful object features.

2.4 Causes

- A very complex model with many parameters.
- A small or insufficiently diverse training dataset.
- Noisy data or incorrectly labelled examples.
- Training for too many epochs.
- No regularization or inadequate validation.

2.5 Remedies

- Use L1 or L2 regularization.

- Use data augmentation.
- Use more diverse data.
- Use early stopping or a smaller model.
- Use labelled and unlabelled data through semi-supervised learning where appropriate.

3. Regularization

3.1 Definition

Regularization is a method for reducing overfitting by adding a penalty or constraint to the original learning objective. The model is encouraged not only to fit the training data but also to keep its parameters controlled.

$$J(W) = L(W) + \lambda R(W)$$

Here, **J(W)** is the total regularized objective, **L(W)** is the original data loss, **R(W)** is the regularization penalty, **W** represents the model weights, and **λ** controls the strength of the penalty.

3.2 Interpretation

- Minimize prediction error on training data.
- Avoid unnecessarily large or overly complex parameter values.
- Improve performance on unseen data, also called generalization.

3.3 Role of λ

Value of λ	Effect
Very small	Regularization is weak; overfitting may remain.
Suitable/moderate	Balances data fitting and weight control; usually improves generalization.
Very large	Penalty dominates; the model may become too simple and underfit.

In practice, λ is selected using a validation set or model-selection procedure.

4. Parameter Norm Penalties

A parameter norm penalty adds a cost based on the magnitude of model weights. The two important penalties are **L1 regularization** and **L2 regularization**.

5. L1 Regularization

5.1 Formula

$$J(W) = L(W) + \lambda \Sigma|W_i|$$

L1 regularization adds the absolute values of all weights to the original loss. It tends to make some less-important weights exactly zero, producing a **sparse model**.

5.2 Main characteristics

- Uses the absolute value of each weight.
- Encourages sparsity: some weights can become exactly zero.
- Can act like feature selection by removing less-useful inputs or connections.
- Useful when a simpler model with fewer active parameters is desired.

5.3 Solved Numerical 1: L1 regularized loss

Given: $W = [2, -3, 4]$, $\lambda = 0.1$, and original loss $L(W) = 2.5$.

Step 1: Calculate the L1 norm

$$\Sigma|W_i| = |2| + |-3| + |4| = 2 + 3 + 4 = 9$$

Step 2: Calculate the penalty

$$\lambda \Sigma|W_i| = 0.1 \times 9 = 0.9$$

Step 3: Calculate regularized objective

$$J(W) = 2.5 + 0.9 = 3.4$$

Answer: L1 norm = 9, L1 penalty = 0.9, and regularized loss = **3.4**.

5.4 Solved Numerical 2: L1 regularized loss

Given: $W = [0.5, -1.5, 2]$, $\lambda = 0.2$, $L(W) = 4$.

$$\Sigma|W_i| = 0.5 + 1.5 + 2 = 4$$

$$\text{Penalty} = 0.2 \times 4 = 0.8$$

$$J(W) = 4 + 0.8 = 4.8$$

Answer: Regularized loss = **4.8**.

5.5 Gradient idea

For a nonzero weight, the derivative of $|W_i|$ is $+1$ for positive W_i and -1 for negative W_i. Conceptually:

$$\nabla J(W) = \nabla L(W) + \lambda \, \text{sign}(W)$$

At zero, the absolute-value function is not ordinarily differentiable; subgradient methods are used in optimization.

Memory aid: L1 → absolute values → sparse or zero weights.

6. L2 Regularization

6.1 Formula

$$J(W) = L(W) + (\lambda/2) \Sigma W_i^2$$

L2 regularization adds the squared values of weights to the loss. It strongly penalizes large weights and encourages all weights to become small, although it usually does not make many weights exactly zero.

6.2 Main characteristics

- Uses squares of weight values.
- Penalizes large weights more strongly because the penalty grows quadratically.
- Produces smoother models with controlled weights.
- Often called weight decay in neural-network training.

6.3 Solved Numerical 1: L2 regularized loss

Given: $W = [1, -2, 3]$, $\lambda = 0.1$, and original loss $L(W) = 2.5$.

Step 1: Calculate sum of squared weights

$$\Sigma W_i^2 = 1^2 + (-2)^2 + 3^2 = 1 + 4 + 9 = 14$$

Step 2: Calculate L2 penalty

$$(\lambda/2) \Sigma W_i^2 = (0.1/2) \times 14 = 0.05 \times 14 = 0.7$$

Step 3: Calculate regularized objective

$$J(W) = 2.5 + 0.7 = 3.2$$

Answer: Sum of squares = 14, L2 penalty = 0.7, and regularized loss = **3.2**.

6.4 Solved Numerical 2: L2 regularized loss

Given: $W = [2, -1, 0.5]$, $\lambda = 0.4$, $L(W) = 3.2$.

$$\Sigma W_i^2 = 2^2 + (-1)^2 + 0.5^2 = 4 + 1 + 0.25 = 5.25$$

$$\text{Penalty} = (0.4/2) \times 5.25 = 0.2 \times 5.25 = 1.05$$

$$J(W) = 3.2 + 1.05 = 4.25$$

Answer: Regularized loss = **4.25**.

6.5 L2 gradient

The derivative of the L2 penalty is λW. Hence:

$$\nabla J(W) = \nabla L(W) + \lambda W$$

The added λW term continuously pulls weights toward zero.

Memory aid: L2 → squared weights → smaller weights.

7. Comparison: L1 vs L2

Aspect	L1 Regularization	L2 Regularization
Formula	$L(W) + \lambda \Sigma \lvert W_i \rvert$	$L(W) + (\lambda/2)\Sigma W_i^2$
Penalty type	Absolute value	Squared value
Effect on weights	Some weights can become exactly zero	Weights are smoothly shrunk toward zero
Model property	Sparse model	Smooth, controlled model
Feature selection	Strong	Generally weak
Large weights	Penalty rises linearly	Penalty rises quadratically
Typical purpose	Discard less-important weights/features	Control large weights and improve generalization

Example: for $W = [0.1, 2, 5]$, L1 norm = 7.1 whereas $\Sigma W_i^2 = 29.01$. This shows why L2 penalizes a large weight such as 5 much more strongly.

8. Numericals on λ

8.1 Solved Numerical: Find λ for L1

Given: $L(W) = 5$, $\Sigma \lvert W_i \rvert = 8$, and $J(W) = 5.8$. Find λ.

$$J(W) = L(W) + \lambda \Sigma \lvert W_i \rvert$$

$$5.8 = 5 + 8\lambda$$

$$0.8 = 8\lambda$$

$$\lambda = 0.8/8 = 0.1$$

Answer: λ = **0.1**.

8.2 Solved Numerical: Find λ for L2

Given: $L(W) = 3$, $\Sigma W_i^2 = 20$, and $J(W) = 4$. Find λ.

$$4 = 3 + (\lambda/2) \times 20$$

$$4 = 3 + 10\lambda$$

$$1 = 10\lambda$$

$$\lambda = 0.1$$

Answer: λ = **0.1**.

9. Weight Decay

9.1 Definition

Weight decay means gradual reduction of weights during learning. It is associated with L2 regularization. Large weights are discouraged because they increase the regularization penalty.

$$W_{new} = W - \eta(\nabla L(W) + \lambda W)$$

Here, η is the learning rate, $\nabla L(W)$ is the original-loss gradient, and λW is the L2 regularization gradient.

9.2 Solved Numerical 1: weight decay update

Given: $W = 2$, $\nabla L(W) = 0.6$, $\lambda = 0.1$, $\eta = 0.05$.

$$\lambda W = 0.1 \times 2 = 0.2$$

$$\text{Total gradient} = 0.6 + 0.2 = 0.8$$

$$W_{new} = 2 - 0.05 \times 0.8 = 2 - 0.04 = 1.96$$

Answer: New weight = **1.96**.

9.3 Solved Numerical 2: negative weight

Given: $W = -3$, $\nabla L(W) = -0.4$, $\lambda = 0.2$, $\eta = 0.1$.

$$\lambda W = 0.2 \times (-3) = -0.6$$

$$\text{Total gradient} = -0.4 + (-0.6) = -1.0$$

$$W_{new} = -3 - 0.1(-1.0) = -3 + 0.1 = -2.9$$

Answer: New weight = **−2.9**. The weight moves closer to zero.

10. Data Augmentation

10.1 Definition

Data augmentation creates additional training variations from existing examples while preserving their correct class labels. It increases the variety available to the model and reduces dependence on exact training samples.

10.2 Image augmentation examples

- Small rotation.
- Horizontal flipping, where class meaning is preserved.
- Cropping and resizing.
- Zooming, translation, or scaling.
- Brightness or contrast change.
- Adding small amounts of realistic noise.

10.3 How it prevents overfitting

A model that sees many realistic variations cannot simply memorize exact pixel patterns or fixed positions. It is encouraged to learn useful, invariant features. For instance, a cat remains a cat after a small rotation, crop, or brightness change.

10.4 Caution

Every transformation must preserve the label. For example, some flips or rotations can change the meaning of handwritten digits or domain-specific symbols. In medical, satellite, or industrial images, only scientifically valid transformations should be applied.

Data augmentation is often called an **implicit regularizer** because it changes the training data distribution rather than directly adding a penalty term to the loss.

11. Semi-Supervised Learning

11.1 Definition

Semi-supervised learning uses a small amount of labelled data together with a larger amount of unlabelled data.

Example: 1,000 labelled student images and 9,000 unlabelled student images.

11.2 Comparison

Approach	Labelled data	Unlabelled data
Supervised learning	Used	Usually not required for the main training objective
Unsupervised learning	Not required	Used
Semi-supervised learning	Small amount used	Large amount used

11.3 Basic working idea

- Train an initial model using available labelled samples.
- Use it to make predictions for unlabelled samples.
- Select sufficiently confident predictions as pseudo-labels, or require prediction consistency under small input perturbations.
- Retrain using labelled examples and carefully selected unlabelled examples.

11.4 Benefits and risk

- Reduces the effort and cost of manual annotation.
- Can improve generalization when labelled data is scarce.
- Incorrect pseudo-labels can reinforce errors; confidence thresholds and validation are important.

12. Optimization in Deep Learning

12.1 Objective

Training a neural network means finding parameter values that minimize a loss or objective function.

$$W^* = \arg\min_W J(W)$$

W* denotes the weight vector that gives the minimum objective value found during training.

12.2 Backpropagation and optimization

- **Backpropagation** computes gradients: how the loss changes with respect to each weight.
- **Optimization** uses those gradients to update the weights and reduce loss.

13. Gradient Descent

13.1 Update rule

$$W_{new} = W_{old} - \eta \nabla J(W)$$

The gradient points in the direction of maximum increase of loss. Subtracting the gradient moves the weights in a direction that typically reduces loss. η is the learning rate, which controls step size.

13.2 Solved Numerical 1

Given: $W = 4$, $\nabla J(W) = 1.5$, $\eta = 0.1$.

$$W_{new} = 4 - 0.1(1.5) = 4 - 0.15 = 3.85$$

Answer: W_{new} = **3.85**.

13.3 Solved Numerical 2

Given: $W = -2$, $\nabla J(W) = -0.8$, $\eta = 0.25$.

$$W_{new} = -2 - 0.25(-0.8) = -2 + 0.2 = -1.8$$

Answer: W_{new} = **−1.8**.

13.4 Solved Numerical 3: vector update

Given: $W = [2, -1, 0.5]$, $\nabla J(W) = [0.4, -0.6, 0.2]$, $\eta = 0.1$.

$$\eta \nabla J(W) = 0.1[0.4, -0.6, 0.2] = [0.04, -0.06, 0.02]$$

$$W_{new} = [2, -1, 0.5] - [0.04, -0.06, 0.02]$$

$$W_{new} = [1.96, -0.94, 0.48]$$

Answer: W_{new} = **[1.96, −0.94, 0.48]**.

14. Challenges in Neural Network Optimization

Deep neural-network loss surfaces are high-dimensional and non-convex. The following syllabus-prescribed conditions can make gradient-based training slow or difficult.

14.1 Ill-Conditioning

Ill-conditioning occurs when the loss surface has very different curvature in different directions. A contour plot would look like a long, narrow valley: very steep across the valley and very flat along it.

- Steep direction: gradients can be large, causing overshooting or oscillation.
- Flat direction: gradients can be small, causing very slow progress.
- A single learning rate may be too large for one direction and too small for another.

Illustrative function:

$$J(x, y) = 10x^2 + 0.5y^2$$

$$\partial J/\partial x = 20x, \partial J/\partial y = y$$

At (x, y) = (1, 1), the gradient is [20, 1], showing that the x-direction changes far more rapidly than the y-direction.

Solved update: Let $\eta = 0.04$.

$$x_{new} = 1 - 0.04(20) = 0.2$$

$$y_{new} = 1 - 0.04(1) = 0.96$$

The x-coordinate changes rapidly, while the y-coordinate changes slowly. This uneven movement illustrates ill-conditioning.

Exam statement: Ill-conditioning is a condition in which the loss function has widely different curvature in different directions. It can cause oscillation in steep directions and slow convergence in flat directions.

14.2 Local Minima

A **local minimum** is a point whose loss is lower than that of nearby points, but it may not have the lowest loss over the entire parameter space. The lowest point overall is the **global minimum**.

Term	Meaning
Local minimum	A low point relative to its neighbouring region.
Global minimum	The lowest objective value across the full parameter space.

If an optimizer reaches a local valley, the local gradient may be zero, so it can stop there even when another region has a smaller loss.

Modern perspective: In very high-dimensional networks, poor local minima are often less important in practice than saddle points and flat regions. Nevertheless, local minima remain an important optimization challenge and syllabus topic.

14.3 Plateaus

A **plateau** is a broad region of the loss surface where gradients are very small. The loss changes little as parameters move through the region.

$$\text{If } \nabla J(W) \approx 0, \text{ then } W_{new} = W - \eta \nabla J(W) \approx W$$

Solved Numerical: Given $W = 2$, $\nabla J(W) = 0.001$, and $\eta = 0.1$:

$$W_{new} = 2 - 0.1(0.001) = 2 - 0.0001 = 1.9999$$

The update is extremely small, so learning is very slow.

- Plateaus may occur because of saturated activation functions, poor initialization, or vanishing gradients.
- They make training appear stalled even though the model may not be at a true minimum.

14.4 Saddle Points

A **saddle point** is a stationary point where the surface curves upward in some directions and downward in others. It is neither a conventional local minimum nor a conventional local maximum.

Classic example:

$$J(x, y) = x^2 - y^2$$

$$\partial J/\partial x = 2x, \partial J/\partial y = -2y$$

At (0, 0), the gradient is zero. However, along the x direction, $J(x, 0) = x^2$ rises away from zero, whereas along the y direction, $J(0, y) = -y^2$ falls away from zero. Thus, (0, 0) is a saddle point.

- Near a saddle point, gradients can be zero or very small.
- Gradient-based training can slow down or remain in the neighbourhood for many iterations.
- In high-dimensional neural networks, saddle points can be more common than bad local minima.

14.5 Flat Regions

A **flat region** is an area where the loss changes very little with changes in parameters. The gradients are often small, so updates are small and convergence is slow.

Plateaus and flat regions are closely related in introductory deep-learning discussions. A plateau is commonly described as a broad flat region; both lead to slow learning because the gradient signal is weak.

15. Summary Table: Optimization Challenges

Challenge	Core idea	Effect on gradient descent
Ill-conditioning	Very different curvature across directions	Oscillations in steep directions; slow progress in flat directions
Local minimum	Lower than nearby points but may not be global	Optimizer can settle in a local valley
Plateau	Broad region with near-zero gradient	Very small updates and slow learning
Saddle point	Upward curvature in some directions and downward in others	Near-zero gradient can slow progress
Flat region	Little change in loss	Small gradients and small parameter updates

16. Integrated Solved Numerical

Problem: A network has $W = [2, -1, 0.5]$, original loss $L(W) = 3$, $\lambda = 0.2$, $\nabla L(W) = [0.4, -0.6, 0.2]$, and learning rate $\eta = 0.1$. Find: (1) L1 regularized objective, (2) L2 regularized objective, and (3) updated weights using L2 regularization.

16.1 L1 objective

$$\Sigma|W_i| = |2| + |-1| + |0.5| = 3.5$$

$$\text{L1 penalty} = 0.2 \times 3.5 = 0.7$$

$$J_{L1} = 3 + 0.7 = 3.7$$

Answer: L1 regularized objective = **3.7**.

16.2 L2 objective

$$\Sigma W_i^2 = 2^2 + (-1)^2 + 0.5^2 = 4 + 1 + 0.25 = 5.25$$

$$\text{L2 penalty} = (0.2/2) \times 5.25 = 0.525$$

$$J_{L2} = 3 + 0.525 = 3.525$$

Answer: L2 regularized objective = **3.525**.

16.3 L2 weight update

$$\lambda W = 0.2[2, -1, 0.5] = [0.4, -0.2, 0.1]$$

$$\nabla J(W) = \nabla L(W) + \lambda W = [0.4, -0.6, 0.2] + [0.4, -0.2, 0.1]$$

$$\nabla J(W) = [0.8, -0.8, 0.3]$$

$$W_{new} = W - \eta \nabla J(W)$$

$$W_{new} = [2, -1, 0.5] - 0.1[0.8, -0.8, 0.3]$$

$$W_{new} = [2, -1, 0.5] - [0.08, -0.08, 0.03]$$

$$W_{new} = [1.92, -0.92, 0.47]$$

Answer: Updated weights = **[1.92, −0.92, 0.47]**.

17. Important Formula Sheet

Topic	Formula
General regularization	$J(W) = L(W) + \lambda R(W)$
L1 regularization	$J(W) = L(W) + \lambda \Sigma \lvert W_i \rvert$
L2 regularization	$J(W) = L(W) + (\lambda/2)\Sigma W_i^2$
Optimization objective	$W^* = \arg\min_W J(W)$
Gradient descent	$W_{new} = W_{old} - \eta \nabla J(W)$
L2 gradient	$\nabla J(W) = \nabla L(W) + \lambda W$
Weight decay update	$W_{new} = W - \eta(\nabla L(W) + \lambda W)$

18. Exam-Focused Questions

- Explain overfitting and regularization.
- Explain parameter norm penalties.
- Explain L1 and L2 regularization with formulations and examples.
- Compare L1 and L2 regularization.
- Explain the role of λ in regularization.
- Explain weight decay with a numerical example.
- Explain data augmentation and how it reduces overfitting.
- Explain semi-supervised learning with a suitable example.
- Explain gradient descent with a numerical example.
- Explain ill-conditioning, local minima, plateaus, saddle points, and flat regions.
- Solve numericals based on L1, L2, regularized objective, λ, gradient descent, and weight decay.

19. Quick Revision Sheet

- **Overfitting:** Good training performance but poor performance on unseen data.
- **Regularization:** Add penalty to reduce overfitting: $J = L + \lambda R$.
- **L1:** Absolute values; encourages sparse or zero weights.
- **L2:** Squared weights; shrinks weights smoothly; related to weight decay.
- **Data augmentation:** Create label-preserving training variations.
- **Semi-supervised learning:** Use a small labelled dataset plus a large unlabelled dataset.
- **Gradient descent:** $W_{new} = W_{old} - \eta \nabla J(W)$.
- **Ill-conditioning:** Different curvature in different directions.
- **Local minimum:** Lower than nearby points but not necessarily globally lowest.
- **Plateau/flat region:** Very small gradient; slow updates.
- **Saddle point:** Minimum-like in some directions and maximum-like in others.

Prepared as expanded study material based on the supplied BCA701 Module 3 notes. The explanations and solved numericals elaborate the listed syllabus topics without adding a separate optimization-algorithms unit.